

EPIS Topic Gap Analyzer

v1.1 Explainer and Output Guide

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Updated: 2026-05-30

Public HTML report: <https://jrivadeneira1.github.io/epis-topic-gap-report/>

106 PDFs indexed	15 Working groups	11 Topic categories	35 Relevant gaps
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Scope of v1.1: The prototype now covers both EPIS geographic working groups and thematic working groups. The report is published online and structured for internal demonstration, not as a final editorial judgment.

1. What I built

I built a working prototype of an EPIS Topic Gap Analyzer. The tool collects EPIS-hosted publication PDFs, extracts the text, scores the content across predefined topic categories, filters candidate gaps through a working-group relevance matrix, and generates both CSV outputs and a readable HTML report.

Version 1.1 expands the first version from geographic working groups only to a broader structure covering 8 geographic working groups and 7 thematic working groups. This makes the output more aligned with how EPIS organizes its FASP content and more useful for topic planning across divisions.

Core purpose: decision support. The tool does not replace editorial judgment; it helps detect possible under-covered themes and convert them into article ideas that can be manually reviewed before being proposed to EPIS.

Working-group coverage

Geographic working groups	Thematic working groups
Arctic and Space; Eurasia; Europe; Indo-Pacific; Middle East and North Africa; North America; South America; Sub-Saharan Africa.	Artificial Intelligence and Cybersecurity; Climate Policy and Environment; Human Rights and Humanitarian Aid; International Economic Relations; International Relations and Diplomacy; Peacekeeping and Conflict Prevention; Security Policy and Defence.

2. Pipeline structure

The pipeline is modular. Each step creates an output file that can be inspected before the next step runs. This makes the project easier to debug, explain, and improve.

1. **main.py:** crawls EPIS working-group pages and downloads EPIS-hosted PDFs. Output: manifest.csv and /downloads.
2. **build_publication_index.py:** turns the manifest into a structured publication index. Output: publication_index.csv.
3. **extract_pdf_text.py:** extracts readable text from each downloaded PDF. Output: text_index.csv and /texts.
4. **analyze_topic_categories.py:** scores topic-category terms in each document and group. Outputs: document_topic_scores.csv and group_topic_scores.csv.
5. **generate_topic_gap_candidates.py:** compares group scores with corpus averages and flags raw candidate gaps. Output: topic_gap_candidates.csv.
6. **generate_relevant_topic_gaps.py:** keeps only weak or absent topics that are relevant for the group. Output: relevant_topic_gaps.csv.
7. **generate_gap_report.py:** adds priorities, interpretations, and possible article ideas. Output: epis_topic_gap_report.csv.
8. **generate_html_report.py:** creates the readable HTML report used for GitHub Pages. Output: reports/epis_topic_gap_report.html.

3. Main measures in the CSV report

The final CSV file is not meant to be read as raw statistics only. Each measure supports a specific interpretation. I use the scores to identify candidate gaps, then review them manually before treating them as useful article ideas.

- **group:** machine-readable group identifier used to separate geographic and thematic divisions.
- **group_label:** readable EPIS group name, for example South America or Security Policy and Defence.
- **topic_label:** broad theme being evaluated, such as Energy security, China's role, or Technology/cyber/AI.
- **coverage_level:** gap means zero detected coverage; weak means clearly below the corpus average.
- **priority:** simplified ranking: High, Medium, or Low.
- **group_score_per_1000_words:** topic frequency inside the group, normalized per 1,000 words.
- **corpus_average_per_1000_words:** benchmark topic frequency across the whole EPIS PDF corpus.
- **gap_priority_score:** distance between corpus average and group score when the group is below average.
- **document_count:** number of EPIS PDFs available for that group in the corpus.
- **interpretation:** plain-language explanation of the result.
- **possible_article_idea:** suggested article angle based on the detected candidate gap.

4. Methodological decisions

- **Topic dictionaries instead of a black-box model:** this keeps the first version transparent and easy to explain.
- **Normalization per 1,000 words:** groups do not have the same number of PDFs or amount of text, so normalized frequencies are more comparable.
- **Relevance matrix by group:** a low score is not automatically a gap. The matrix prevents irrelevant false positives.
- **Separate geographic and thematic divisions:** the HTML report now reflects the EPIS content architecture instead of mixing every group alphabetically.
- **Human review remains central:** the tool identifies candidate gaps; a researcher still decides whether a suggestion is intellectually useful and suitable for EPIS.

5. How I interpret the results

I read the results as a shortlist of possible article directions, not as final conclusions. A high-priority candidate means that a topic is both relevant to the working group and weakly represented in the current PDF corpus according to the dictionary-based scoring.

For example, if South America shows a weak score for migration or China's role, I do not treat that as automatic proof that EPIS has ignored the topic. I treat it as a prompt for manual review: I check the titles, compare the group mandate, and decide whether a new 500-word contribution would add value.

Important limitation: The prototype depends on the quality of the keyword dictionaries and on the available EPIS-hosted PDFs. Some EPIS publications may exist as web pages without PDFs, and those are not fully represented in the current text corpus. The current version should therefore be treated as a transparent prototype, not as a complete map of all EPIS content.

6. Final deliverables produced by v1.1

- **epis_topic_gap_report.csv:** structured report with candidate gaps, scores, interpretations, and article ideas.
- **reports/epis_topic_gap_report.html:** readable report suitable for internal demonstration and GitHub Pages publication.
- **GitHub Pages public report:** browser-accessible version of the HTML report.
- **EPIS_Topic_Gap_Analyzer_v1_1_Guide.pdf:** short documentation note explaining what I built and how to interpret the measures.